



Firewall Rulebase Governance and Rule Register

Purpose: The audit-facing record of the pfSense firewall rulebase. It states the principles the rulebase is built on, records every rule with its business justification and the control it satisfies, and logs every change. It is written so that an auditor (or a new engineer) can understand *why* each rule exists without asking anyone.

Why this document exists: A firewall configuration on its own is not auditable. Auditors ask three questions of every rule: *why does it exist, who approved it, and is it still needed?* This register answers all three in one place. The discipline mirrors **PCI-DSS Requirement 1** (documented ruleset with justification and periodic review), applied here to a NIST 800-53 / CIS Controls context.

Status: Living document. Updated every time a rule is added, changed, or removed.

1. Rulebase principles

#	Principle	What it means in practice	Framework
1	Deny by default	Nothing passes unless a rule allows it. Every interface ends in an implicit deny.	NIST SC-7(5), CIS 4.4
2	Least privilege	Every rule is the narrowest possible: specific source, destination, service. No any → any .	NIST CM-7, AC-6, CIS 4.8
3	Documented justification	Every rule has a recorded business reason, owner, and date (this register).	NIST CM-2/CM-6, PCI-DSS 1.1
4	Correct ordering	Rules are ordered so specific precedes broad; no shadowed or redundant rules.	NIST CM-6
5	Logging & accountability	Denies and sensitive allows are logged to the SIEM (Wazuh).	NIST AU-2/AU-3
6	Change control	Changes are proposed, approved, dated, reversible (disable before delete).	NIST CM-3
7	Periodic recertification	The rulebase is reviewed on a fixed cadence; stale rules removed.	NIST CA-7, PCI-DSS 1.1.7
8	Boundary enforcement	Rules enforce the VLAN/zone boundaries defined by segmentation.	NIST SC-7, CIS 12

2. Rule register

Each rule carries a stable **Rule ID** (<TAB>-<NN>) that does not change even if the rule's position moves. "Log" = whether the rule logs matches to the SIEM. "Status" = Built or Planned.

2.1 MANAGEMENT interface (VLAN 10, source = admin/control plane)

Rule ID	Source	Destination	Service	Action	Log	Justification	Control	Status
MGMT-01	MANAGEMENT net	This Firewall	PFSense_MGMT (443,80,22)	Pass	Yes	Anti-lockout: guarantee admin access to pfSense from the management plane before broad rules are removed. Admin access is logged for accountability.	AC-17, CM-3, AU-2	Built
MGMT-02	MANAGEMENT net	This Firewall	TCP/UDP 53	Pass	No	DNS resolution via the pfSense resolver for management hosts.	SC-7	Built
MGMT-03	MANAGEMENT net	This Firewall	UDP 123	Pass	No	Time sync (NTP) from pfSense; correct time is required for logging and Kerberos.	AU-8	Built
MGMT-04	MANAGEMENT net	any	ICMP echo-request	Pass	No	Diagnostic reachability (ping) for troubleshooting. Echo-request only (least privilege); replies return via stateful inspection.	CA-7	Built
MGMT-05	MANAGEMENT net	SRV1_DC	AD_TCP	Pass	No	Domain membership (TCP) for management-plane Windows hosts joining ad.biira.online .	AC-4	Built
MGMT-06	MANAGEMENT net	SRV1_DC	AD_UDP	Pass	No	Domain membership (UDP).	AC-4	Built
MGMT-07	MANAGEMENT net	SRV1_DC	AD_RPC_DYNAMIC	Pass	No	RPC high ports for domain join, Group Policy, replication.	AC-4	Built
MGMT-08	MANAGEMENT net	SRV1_DC	MGMT_TCP	Pass	Yes	Administrative access (RDP, WinRM/Ansible) to the DC. Logged as privileged access.	AC-6, AC-17, AU-2	Built

Rule ID	Source	Destination	Service	Action	Log	Justification	Control	Status
MGMT-09	MANAGEMENT net	LAB_NETS	any	Pass	No	Administrative reach into all lab VLANs from the management plane (explicit; to be tightened per-service over time).	AC-6	Built
MGMT-10	MANAGEMENT net	NOT LAB_NETS	any	Pass	No	Internet access for the management plane (destination inverted = anywhere except the lab).	SC-7	Built
MGMT-11	MANAGEMENT net	SIEM01_HOST	SIEM_ADMIN (22, 443)	Pass	Yes	SSH and dashboard access to SIEM01 for administration. Logged, because administrative access to the host holding the security evidence must be attributable.	AC-17, AU-2	Built
(implicit)	any	any	any	Deny	-	Default deny. Anything not explicitly allowed is dropped.	SC-7(5)	Built-in

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🔒 Not secure https://192.168.10.1/firewall_rules.php?if=opt1

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Relaunch to update

Firewall / Rules / MANAGEMENT

📄📑?

The changes have been applied successfully. The firewall rules are now reloading in the background.
Monitor the filter reload progress.

Floating

Tailscale

WAN

LAN

MANAGEMENT

BLUETEAM

REDTEAM

DEVOPS

ENTERPRISELAN

MONITORING

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ ☰	0/0 B	IPv4 TCP	MANAGEMENT subnets	*	This Firewall (self)	PFSense_MGMT	*	none	MGMT-01 - Anti-lockout: MGMT VLAN admin access to pfSense (web UI/SSH). NIST AC-17/CM-3/AU-2	
☐	✓	0/2.33 MiB	IPv4 *	MANAGEMENT subnets	*	BLUETEAM subnets	*	*	none	Allow Management to reach BlueTeam	
☐	✓	0/0 B	IPv4 TCP	MANAGEMENT subnets	*	192.168.50.2	3389 (MS RDP)	*	none	Allow RDP from Management to Windows Server 2022	
☐	✓	0/881 KiB	IPv4 TCP/UDP	MANAGEMENT subnets	*	This Firewall (self)	53 (DNS)	*	none	Allow MGMT DNS to pfSense	
☐	✓	0/1.79 GiB	IPv4 *	*	*	*	*	*	none	All	
☐	✓	0/0 B	IPv4 *	MANAGEMENT subnets	*	MONITORING subnets	*	*	none	Reach Monitoring VLAN on Any Protocol	
☐	✓	0/0 B	IPv4 UDP	MANAGEMENT subnets	*	*	*	*	none	Allow DNS lookups	
☐	✓	0/0 B	IPv4 ICMP any	MANAGEMENT subnets	*	*	*	*	none	Allow ping to pfSense from VLAN 10	
☐	✓	0/0 B	IPv4 TCP	*	*	This Firewall (self)	*	*	none	Allow Web UI from VLAN 10	

Figure 13.1: The MANAGEMENT tab partway through the build. The broad **ALL** rule is still present near the bottom, which is why the specific rules above it could be added safely: nothing was cut off while the ruleset was incomplete.

System Interfaces Firewall Services VPN Status Diagnostics Help 45																																																																																																																																															
Firewall / Rules / MANAGEMENT Floating Tailscale WAN LAN MANAGEMENT BLUETEAM REDTEAM DEVOPS ENTERPRISELAN MONITORING																																																																																																																																															
Rules (Drag to Change Order) <table> <tr> <th></th><th>States</th><th>Protocol</th><th>Source</th><th>Port</th><th>Destination</th><th>Port</th><th>Gateway</th><th>Queue</th><th>Schedule</th><th>Description</th><th>Actions</th></tr> <tr> <td>☐</td><td></td><td>0/0 B</td><td>IPv4 TCP</td><td>MANAGEMENT subnets</td><td>*</td><td>This Firewall (self)</td><td>PFSense_MGMT</td><td>*</td><td>none</td><td>MGMT-01 - Anti-lockout: MGMT VLAN admin access to pfSense (web UI/SSH). NIST AC-17/CM-3/AU-2</td><td></td></tr> <tr> <td>☐</td><td></td><td>0/0 B</td><td>IPv4 TCP/UDP</td><td>MANAGEMENT subnets</td><td>*</td><td>This Firewall (self)</td><td>53 (DNS)</td><td>*</td><td>none</td><td>MGMT-02 - DNS resolution via pfSense resolver. NIST SC-7</td><td></td></tr> <tr> <td>☐</td><td></td><td>0/0 B</td><td>IPv4 UDP</td><td>MANAGEMENT subnets</td><td>*</td><td>This Firewall (self)</td><td>123 (NTP)</td><td>*</td><td>none</td><td>MGMT-03 - NTP time sync from pfSense; correct time underpins logging and Kerberos. NIST AU-8</td><td></td></tr> <tr> <td>☐</td><td></td><td>0/0 B</td><td>IPv4 ICMP echoreq</td><td>MANAGEMENT subnets</td><td>*</td><td>*</td><td>*</td><td>*</td><td>none</td><td>MGMT-04 - ICMP echo for troubleshooting reachability. NIST CA-7</td><td></td></tr> <tr> <td>☐</td><td></td><td>0/0 B</td><td>IPv4 TCP</td><td>MANAGEMENT subnets</td><td>*</td><td>192.168.50.2</td><td>3389 (MS RDP)</td><td>*</td><td>none</td><td>Allow RDP from Management to Windows Server 2022</td><td></td></tr> <tr> <td>☐</td><td></td><td>0/5.35 MIB</td><td>IPv4 *</td><td>MANAGEMENT subnets</td><td>*</td><td>BLUETEAM subnets</td><td>*</td><td>*</td><td>none</td><td>Allow Management to reach BlueTeam</td><td></td></tr> <tr> <td>☐</td><td></td><td>0/2.99 MIB</td><td>IPv4 TCP/UDP</td><td>MANAGEMENT subnets</td><td>*</td><td>This Firewall (self)</td><td>53 (DNS)</td><td>*</td><td>none</td><td>Allow MGMT DNS to pfSense</td><td></td></tr> <tr> <td>☐</td><td></td><td>165/5.28 GiB</td><td>IPv4 *</td><td>*</td><td>*</td><td>*</td><td>*</td><td>*</td><td>none</td><td>All</td><td></td></tr> <tr> <td>☐</td><td></td><td>0/0 B</td><td>IPv4 *</td><td>MANAGEMENT subnets</td><td>*</td><td>MONITORING subnets</td><td>*</td><td>*</td><td>none</td><td>Reach Monitoring VLAN on Any Protocol</td><td></td></tr> <tr> <td>☐</td><td></td><td>0/0 B</td><td>IPv4 UDP</td><td>MANAGEMENT subnets</td><td>*</td><td>*</td><td>*</td><td>*</td><td>none</td><td>Allow DNS lookups</td><td></td></tr> </table>													States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions	☐		0/0 B	IPv4 TCP	MANAGEMENT subnets	*	This Firewall (self)	PFSense_MGMT	*	none	MGMT-01 - Anti-lockout: MGMT VLAN admin access to pfSense (web UI/SSH). NIST AC-17/CM-3/AU-2		☐		0/0 B	IPv4 TCP/UDP	MANAGEMENT subnets	*	This Firewall (self)	53 (DNS)	*	none	MGMT-02 - DNS resolution via pfSense resolver. NIST SC-7		☐		0/0 B	IPv4 UDP	MANAGEMENT subnets	*	This Firewall (self)	123 (NTP)	*	none	MGMT-03 - NTP time sync from pfSense; correct time underpins logging and Kerberos. NIST AU-8		☐		0/0 B	IPv4 ICMP echoreq	MANAGEMENT subnets	*	*	*	*	none	MGMT-04 - ICMP echo for troubleshooting reachability. NIST CA-7		☐		0/0 B	IPv4 TCP	MANAGEMENT subnets	*	192.168.50.2	3389 (MS RDP)	*	none	Allow RDP from Management to Windows Server 2022		☐		0/5.35 MIB	IPv4 *	MANAGEMENT subnets	*	BLUETEAM subnets	*	*	none	Allow Management to reach BlueTeam		☐		0/2.99 MIB	IPv4 TCP/UDP	MANAGEMENT subnets	*	This Firewall (self)	53 (DNS)	*	none	Allow MGMT DNS to pfSense		☐		165/5.28 GiB	IPv4 *	*	*	*	*	*	none	All		☐		0/0 B	IPv4 *	MANAGEMENT subnets	*	MONITORING subnets	*	*	none	Reach Monitoring VLAN on Any Protocol		☐		0/0 B	IPv4 UDP	MANAGEMENT subnets	*	*	*	*	none	Allow DNS lookups	
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☐		0/0 B	IPv4 UDP	MANAGEMENT subnets	*	*	*	*	none	Allow DNS lookups																																																																																																																																					

Figure 13.2: The same tab after MGMT-01 to 04 were in place. Several legacy rules with informal descriptions are still visible; they were replaced by identified rules rather than edited, so the register and the live config could be reconciled line by line.

☐		0/0 B	IPv4 TCP	MANAGEMENT subnets	*	SRV1_DC	AD_TCP	*	none	MGMT-05 - AD domain membership (TCP) to SRV1 DC. NIST AC-4	
☐		0/0 B	IPv4 UDP	MANAGEMENT subnets	*	SRV1_DC	AD_UDP	*	none	MGMT-06 - AD domain membership (UDP) to SRV1 DC. NIST AC-4	
☐		0/0 B	IPv4 TCP	MANAGEMENT subnets	*	SRV1_DC	AD_RPC_DYNAMIC	*	none	MGMT-07 - RPC dynamic ports (domain join, GPO, replication) to SRV1 DC. NIST AC-4	
☐		0/0 B	IPv4 TCP	MANAGEMENT subnets	*	SRV1_DC	MGMT_TCP	*	none	MGMT-08 - Admin access (RDP + WinRM/Ansible) to SRV1 DC. NIST AC-6/AC-17/AU-2	

Figure 13.3: MGMT-05 to 08, the four rules that give the management plane access to the domain controller. Each uses an alias rather than a literal address or port list, so the rule reads as its intent and a change to the alias updates every rule at once.

6/AC-17/AU-2											
☐		0/0 B	IPv4 *	MANAGEMENT subnets	*	LAB_NETS	*	*	none	MGMT-09 - Admin reach into all lab VLANs. NIST AC-6	
☐		0/0 B	IPv4 *	MANAGEMENT subnets	*	! LAB_NETS	*	*	none	MGMT-10 - Internet access (anywhere except the lab). NIST SC-7	
☐		0/0 B	IPv4 TCP	MANAGEMENT subnets	*	192.168.50.2	3389 (MS RDP)	*	none	Allow RDP from Management to Windows Server 2022	

Figure 13.4: MGMT-09 and MGMT-10, the pair that closes the interface. MGMT-09 permits administrative reach into the lab VLANs; MGMT-10 uses an inverted destination, "not LAB_NETS", to grant internet access without granting anything internal. One inverted rule replaces a fragile stack of blocks.

2.2 ENTERPRISELAN interface (VLAN 50, source = the domain controller's segment)

This tab governs what VLAN-50 hosts (currently only `dc01`) may *initiate*. Inbound access to the DC is governed by the source VLANs' tabs (e.g. VLAN 10 via MGMT-05–08) and replies are stateful, so here we permit only what the DC legitimately originates and deny lateral movement into other VLANs.

Rule ID	Source	Destination	Service	Action	Log	Justification	Control	Status
ENT-01	ENTERPRISELAN net	This Firewall	TCP/UDP 53	Pass	No	DNS resolution/forwarding via pfSense.	SC-7	Built

Rule ID	Source	Destination	Service	Action	Log	Justification	Control	Status
ENT-02	ENTERPRISELAN net	This Firewall	UDP 123	Pass	No	NTP time sync; correct time underpins Kerberos and logging.	AU-8	Built
ENT-03	ENTERPRISELAN net	any	ICMP echo-request	Pass	No	Diagnostic reachability (outbound ping).	CA-7	Built
ENT-04	ENTERPRISELAN net	NOT LAB_NETS	any	Pass	No	Internet access (updates, activation, CRL, public DNS). Excludes all lab VLANs.	SC-7	Built
(implicit)	any	any	any	Deny	-	Default deny. VLAN 50 cannot initiate into other VLANs or reach pfSense admin.	SC-7(5), AC-4	Built-in

The `any→any` rule that previously governed this interface was deleted 2026-08-30 after validation.

2.3 REDTEAM interface (VLAN 30, the attack segment)

VLAN 30 is treated as **untrusted**. The baseline contains it from the entire internal lab: the attack box may function (name resolution, time, internet for tooling) but has **no standing path to the domain controller, any other VLAN, or the pfSense admin interface**. Access to a target for an actual exercise is granted as a temporary, specific rule and removed afterwards, so a compromised or careless attack box cannot roam the network.

Rule ID	Source	Destination	Service	Action	Log	Justification	Control	Status
RED-01	REDTEAM net	This Firewall	TCP/UDP 53	Pass	No	Name resolution so the attack host functions.	SC-7	Built
RED-02	REDTEAM net	This Firewall	UDP 123	Pass	No	Time sync; accurate timestamps for exercise evidence.	AU-8	Built
RED-03	REDTEAM net	NOT LAB_NETS	any	Pass	No	Internet only, for tooling and updates. Excludes every lab VLAN.	SC-7	Built
(implicit)	any	any	any	Deny	,	Default deny. No path to DC01, other VLANs, or pfSense admin.	SC-7(5), AC-4	Built-in

Deliberate omission: unlike MANAGEMENT and ENTERPRISELAN, there is **no ICMP-to-any rule** here. An outbound ping rule would let the attack box sweep the internal VLANs for live hosts, which is exactly the reconnaissance step to deny. ICMP to the internet still works through RED-03. The absence of a rule is itself the control.

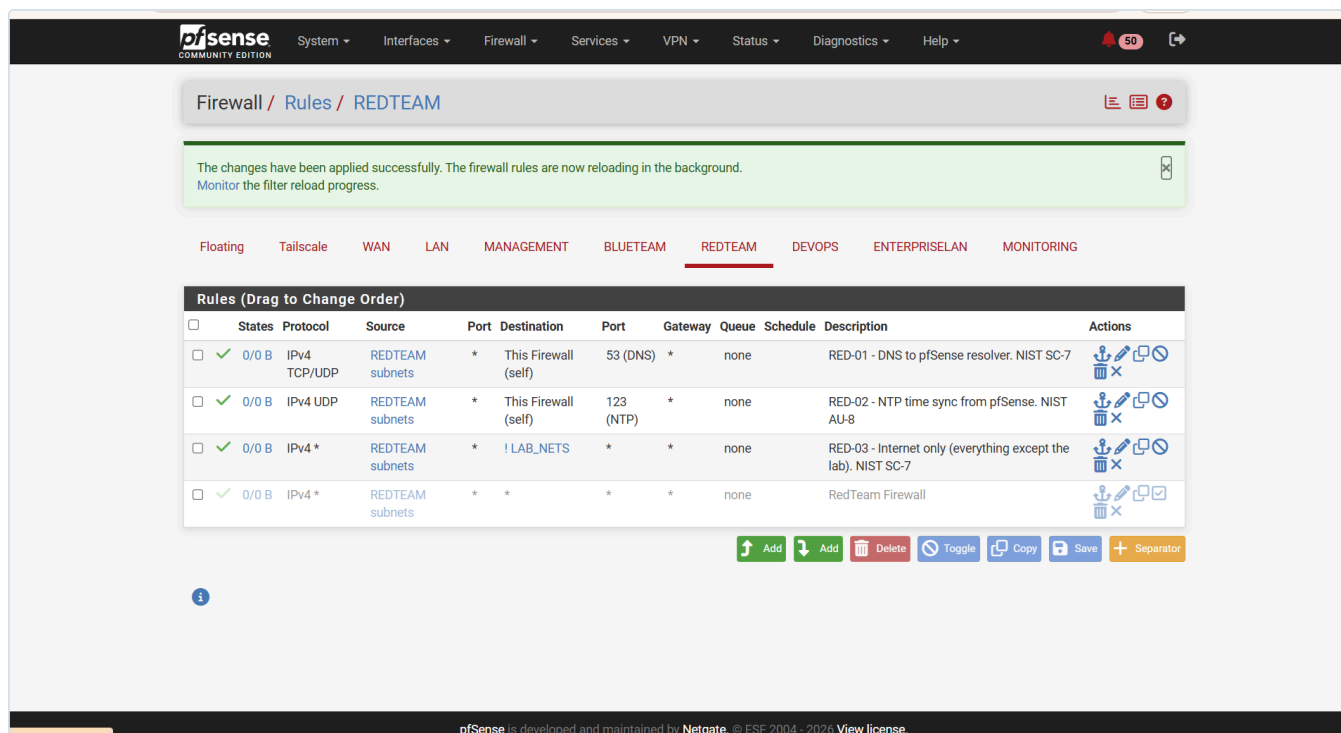


Figure 13.8: The REDTEAM tab. RED-01 to RED-03 are active; the legacy **RedTeam Firewall** any-to-any rule is disabled (greyed) pending deletion. No rule permits VLAN 30 to reach any lab subnet.

Validation status: VALIDATED 2026-09-04. KALI01 (192.168.30.2) was built on VLAN 30 and the containment test was run from it. The attack box resolves names, syncs time and reaches the internet, and cannot reach the domain controller or the firewall's management plane. Full results in section 3.1. The legacy any-to-any rule was deleted the same day, so the interface now carries only RED-01 to RED-03.

Why the denials are timeouts rather than refusals. Both blocked tests returned **Connection timed out**, not **Connection refused**. pfSense drops the packet silently instead of returning a TCP reset or an ICMP unreachable. An attacker on this segment therefore learns nothing from the attempt: a dropped probe is indistinguishable from a host that does not exist, whereas a reset would confirm that something is there and a filter is in front of it. Silent drop is the pfSense default and is retained deliberately.

Remaining tabs (BLUETEAM, DEVOPS, MONITORING) are registered here as they are built. Their target layouts are in `docs/11-domain-controller-firewall.md` section 6.

3. Change control log

Every change to the rulebase is recorded here: what changed, when, who made it, whether it was tested, and how it can be rolled back.

Date	Rule(s)	Change	By	Tested	Rollback
2026-08-15	Aliases	Created <code>SRV1_DC</code> , <code>AD_TCP</code> , <code>AD_UDP</code> , <code>AD_RPC_DYNAMIC</code> , <code>MGMT_TCP</code> , <code>LAB_NETS</code>	Noble Antwi	Verified in Aliases UI	Delete aliases (no rules depend yet)
2026-08-15	MGMT-01	Added anti-lockout rule at top of MANAGEMENT	Noble Antwi	Reload pfSense UI from laptop, confirmed access	Disable rule; broad "All" rule still present

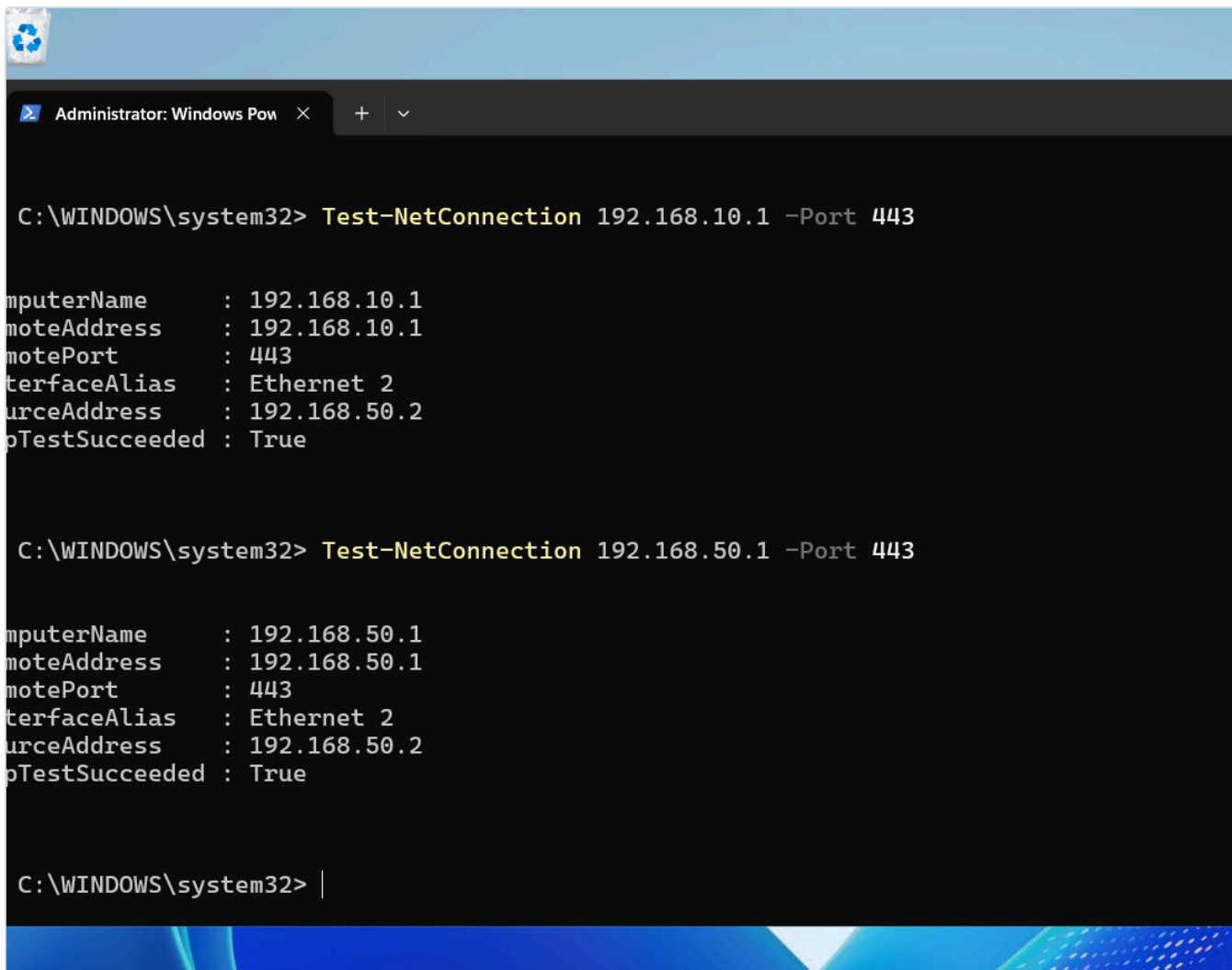
Date	Rule(s)	Change	By	Tested	Rollback
2026-08-15	MGMT-02	Added DNS-to-pfSense rule below MGMT-01	Noble Antwi	UI reachable after apply	Disable rule
2026-08-15	MGMT-03	Added NTP-to-pfSense rule	Noble Antwi	UI reachable after apply	Disable rule
2026-08-15	MGMT-04	Added ICMP rule; corrected subtype echo-reply → echo-request (least privilege, outbound ping)	Noble Antwi	Rule applied	Disable rule
2026-08-15	MGMT-05 to 08	Added the four DC-access rules (AD_TCP, AD_UDP, AD_RPC_DYNAMIC, MGMT_TCP) to SRV1_DC; MGMT-08 logging enabled	Noble Antwi	Aliases resolve as links; applied	Disable rules; DC still reachable via broad "All" until it is removed
2026-08-30	ENT-01 to 04	Built ENTERPRISELAN outbound rules (DNS/NTP to pfSense, ICMP, internet via !LAB_NETS); deleted the prior any→any	Noble Antwi	Isolation test flipped True→False after a manual filter reload (see 3.1 and incident below)	Re-add a temporary any→any on ENTERPRISELAN
2026-09-03	RED-01 to 03	Built REDTEAM baseline (DNS/NTP to pfSense, internet via !LAB_NETS); no ICMP-to-any by design; legacy any-to-any disabled pending deletion	Noble Antwi	Not yet: no host on VLAN 30 (Kali not built). Test deferred, see 2.3	Re-enable the legacy any-to-any rule
2026-09-04	RED-01 to 03	Validated the REDTEAM baseline against a live host (KALI01, 192.168.30.2). No rule changes required; the baseline behaved as designed on first test	Noble Antwi	Yes, three tests, see 3.1	N/A (no change made)
2026-09-04	(legacy)	Deleted the disabled RedTeam Firewall any-to-any rule now that containment is proven. A disabled permit-all left in place is ambiguous to a reviewer	Noble Antwi	Post-deletion tests unchanged	Re-create an any→any pass on REDTEAM
2026-09-04	(service)	Bound the DNS Resolver to the REDTEAM interface (Services → DNS Resolver → Network Interfaces). RED-01 was passing traffic, but unbound was not listening on that interface, so queries timed out	Noble Antwi	nslookup kali.org 192.168.30.1 returns an answer	Deselect REDTEAM in the resolver's interface list
2026-09-04	(service)	Bound the NTP server explicitly to the internal interfaces and excluded WAN . It had been running on the wildcard, which also exposed it on the WAN address	Noble Antwi	nmap -sU -p123 --script ntp-info 192.168.30.1 returns stratum 2; KALI01 clock synchronised	Clear the interface selection to return to the wildcard
2026-08-30	(incident)	Stale filter reload , GUI rule changes were saved but the kernel kept enforcing the old ruleset, so isolation tests kept passing. Root-caused by systematic elimination (floating rules empty, anti-lockout disabled, packet-filtering enabled, any→any deleted, none had effect). Resolved via Status → Filter Reload → Reload Filter , which loaded the current ruleset.	Noble Antwi	Both isolation tests then returned False	N/A (diagnostic)

3.1 Control validation tests

Evidence that a control does what its policy claims. Each test is repeatable, and its result is recorded before and after the change that is meant to affect it.

Date	Control	Test	From → To	Expected	Result
2026-08-15	Management-plane isolation (baseline)	pfSense web UI load	Admin laptop (VLAN 10) → pfSense admin	Reachable	True
2026-08-15	Management-plane isolation (baseline)	Test-NetConnection 192.168.50.1 -Port 443	DC01 (VLAN 50) → pfSense admin 443	Reachable now (VLAN 50 not yet tightened)	True (baseline)
2026-08-30	Management-plane isolation (enforced)	Test-NetConnection 192.168.50.1 -Port 443	DC01 (VLAN 50) → pfSense admin 443	Blocked after ENTERPRISELAN tightened	False ✓
2026-08-30	Management-plane isolation (cross-VLAN)	Test-NetConnection 192.168.10.1 -Port 443	DC01 (VLAN 50) → Management gateway admin 443	Blocked	False ✓
2026-09-04	RedTeam containment (lateral movement)	nc -zvw3 192.168.50.2 445	KALI01 (VLAN 30) → DC01 SMB 445	Blocked	Connection timed out ✓
2026-09-04	RedTeam containment (management plane)	nc -zvw3 192.168.10.1 443	KALI01 (VLAN 30) → Management gateway admin 443	Blocked	Connection timed out ✓
2026-09-04	RedTeam usability (RED-03)	ping -c2 8.8.8.8	KALI01 (VLAN 30) → internet	Reachable	0% packet loss ✓
2026-09-04	RedTeam usability (RED-01)	nslookup kali.org 192.168.30.1	KALI01 (VLAN 30) → pfSense resolver	Resolves	Answer returned ✓
2026-09-04	RedTeam usability (RED-02)	timedatectl after pointing at 192.168.30.1	KALI01 (VLAN 30) → pfSense NTP	Synchronised	System clock synchronized: yes ✓

The change from **True to False**, while `dcdiag`, DNS and internet still pass, is the proof that VLAN 50 is contained: it can use the gateway's DNS/NTP and reach the internet, but cannot reach the pfSense admin UI on any interface, nor pivot into other VLANs. (Ping still succeeds, because ENT-03 permits ICMP, showing the rule is surgical, not a blanket block.)



```
Administrator: Windows PowerShell
C:\WINDOWS\system32> Test-NetConnection 192.168.10.1 -Port 443

ComputerName           : 192.168.10.1
RemoteAddress          : 192.168.10.1
RemotePort             : 443
InterfaceAlias         : Ethernet 2
SourceAddress          : 192.168.50.2
PingTestSucceeded      : True

C:\WINDOWS\system32> Test-NetConnection 192.168.50.1 -Port 443

ComputerName           : 192.168.50.1
RemoteAddress          : 192.168.50.1
RemotePort             : 443
InterfaceAlias         : Ethernet 2
SourceAddress          : 192.168.50.2
PingTestSucceeded      : True

C:\WINDOWS\system32> |
```

Figure 13.5: Before: both `Test-NetConnection ... 443` return True. The DC could reach the pfSense admin UI on its own VLAN and cross-VLAN.

Figure 13.6: After: both return False while Ping stays True. VLAN 50 is denied admin access on every interface, yet the DC's own services keep working.

Operational note, the stale filter reload. The isolation test kept returning True even after the rules were correct, because pfSense had not loaded the new ruleset into the kernel: the GUI and the running filter were out of sync. Systematic elimination (floating rules, anti-lockout, packet-filtering, deleting the rule) confirmed no rule change was taking effect. It was resolved with **Status** → **Filter Reload** → **Reload Filter**. **Lesson:** after tightening rules, if behaviour does not change, verify the filter actually reloaded, a stuck reload silently enforces the old rules and can make a control look broken when it is merely not loaded.

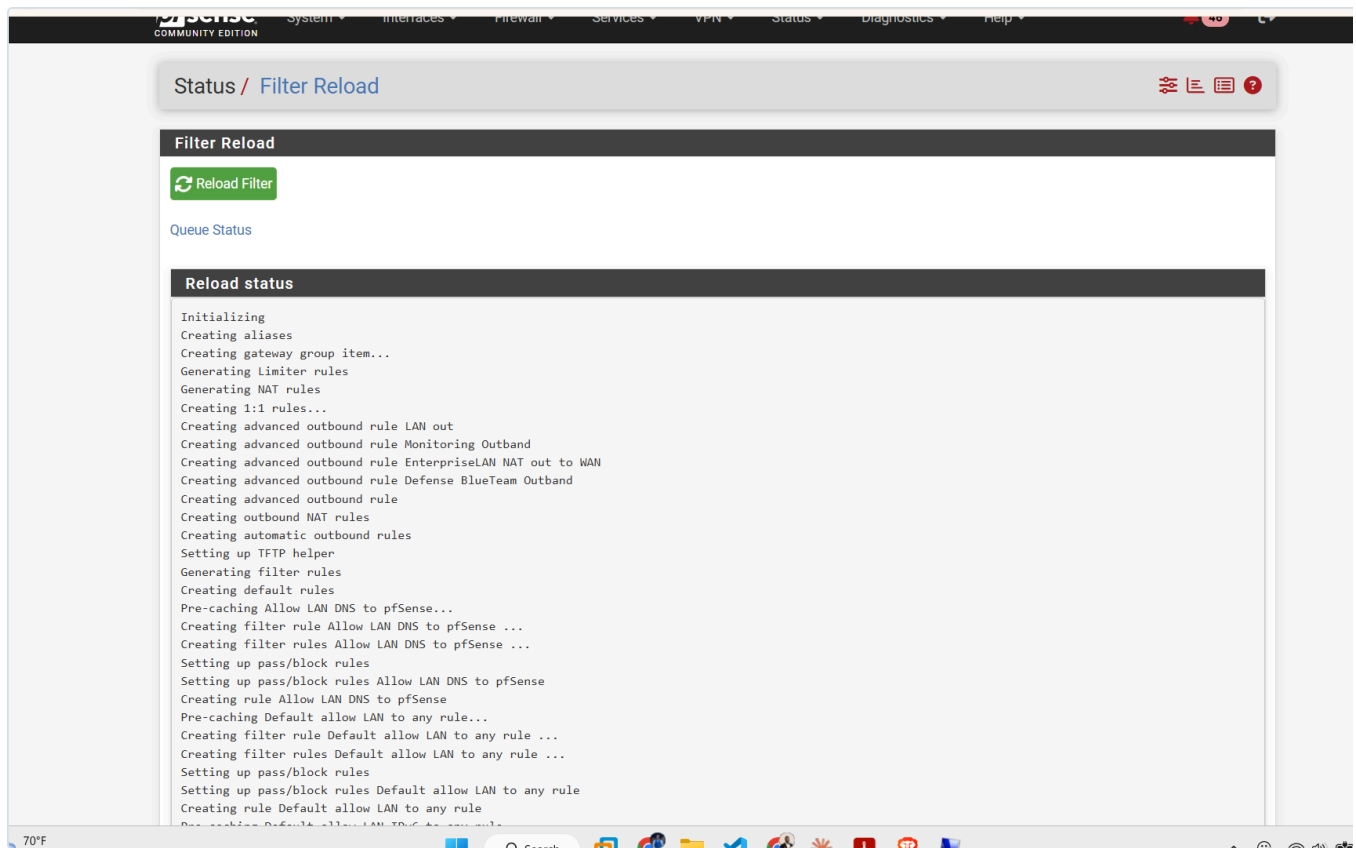


Figure 13.7: Status → Filter Reload. Manually reloading compiled and loaded the current ruleset, after which the isolation control took effect.

```
nantwi@kali01: ~  
[nantwi@kali01]-[~]  
$ nc -zv3 192.168.50.2 445  
192.168.50.2: inverse host lookup failed: Unknown host  
(UNKNOWN) [192.168.50.2] 445 (microsoft-ds) : Connection timed out  
  
[nantwi@kali01]-[~]  
$ nc -zv3 192.168.10.1 443  
192.168.10.1: inverse host lookup failed: Unknown host  
(UNKNOWN) [192.168.10.1] 443 (https) : Connection timed out  
  
[nantwi@kali01]-[~]  
$ ping -c2 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=12.9 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=15.0 ms  
  
--- 8.8.8.8 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1001ms  
rtt min/avg/max/mdev = 12.936/13.965/14.995/1.029 ms  
  
[nantwi@kali01]-[~]  
$ |
```

Figure 13.9: The containment test run from KALI01 on VLAN 30. SMB to DC01 and HTTPS to the management gateway both time out; the internet remains reachable at 0% packet loss. Full build context in [docs/15](#).

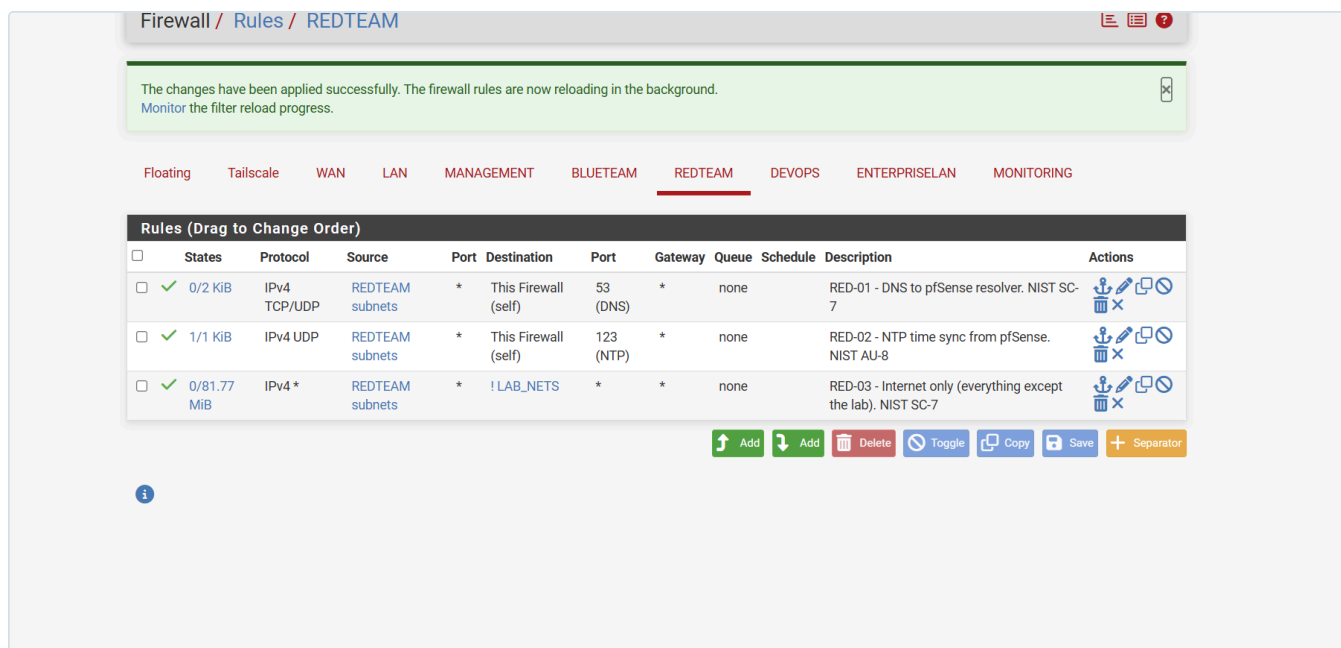


Figure 13.10: The REDTEAM interface after the legacy any-to-any rule was deleted. Three rules, each carrying its identifier, justification and control mapping. Compare Figure 13.8, where the disabled permit-all was still present.

RedTeam containment, validated against a live attack host. The five tests dated 2026-09-04 were run from KALI01 once it was built on VLAN 30. They are deliberately paired: two prove the segment is contained, three prove it is still usable. A containment control that also breaks the host is not a control anyone will keep, so the evidence has to show both halves. The attack box can resolve names, hold accurate time and reach the internet for tooling, while SMB to the domain controller and HTTPS to the firewall's management address both fail.

On the test path. These results measure the rulebase itself. KALI01 runs no VPN client, so its only route is `default via 192.168.30.1`, and every probe was evaluated on the REDTEAM interface. This matters because the administrator's own workstation reaches KALI01 over the Tailscale overlay, a path that does not traverse the per-interface rules at all (see section 6, H-02). The distinction is worth stating plainly: an overlay that terminates inside the perimeter can make a segment look reachable in ways the documented rulebase never authorised. Tests of a segmentation control must originate from a host inside that segment, not from an administrator's machine sitting on an overlay.

A second instance of the config-versus-running-state gap. DNS from VLAN 30 timed out even though RED-01 was demonstrably passing traffic: the rule showed live state entries in the pfSense **States** column. That single observation located the fault. If the firewall had been dropping the queries there would have been no states at all, so the packets were being delivered and nothing was answering. The DNS Resolver was simply not bound to the REDTEAM interface, which had been created after the resolver was first configured. This is the same failure family as the stale filter reload above: the saved configuration was correct, but the running system was not doing what the configuration described.

Lesson: when validating a control, read the state counters before changing any rules. They separate "the firewall blocked it" from "the firewall passed it and the service was not there", which are opposite problems with opposite fixes.

Floating	Tailscale	WAN	LAN	MANAGEMENT	BLUETEAM	REDTEAM	DEVOPS	ENTERPRISELAN	MONITORING
----------	-----------	-----	-----	------------	----------	---------	--------	---------------	------------

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 3/1 KiB	IPv4 TCP/UDP	REDTEAM subnets	*	This Firewall (self)	53 (DNS)	*	none		RED-01 - DNS to pfSense resolver. NIST SC-7	  
☐	✓ 0/0 B	IPv4 UDP	REDTEAM subnets	*	This Firewall (self)	123 (NTP)	*	none		RED-02 - NTP time sync from pfSense. NIST AU-8	  
☐	✓ 0/81.77 MiB	IPv4 *	REDTEAM subnets	*	! LAB_NETS	*	*	none		RED-03 - Internet only (everything except the lab). NIST SC-7	  
☐	✓ 0/0 B	IPv4 *	REDTEAM subnets	*	*	*	*	none		RedTeam Firewall	  

 Add
  Add
  Delete
  Toggle
  Copy
  Save
  Separator

Figure 13.11: RED-01 showing

3/1 KiB in the States column while DNS queries from VLAN 30 were timing out. A blocked packet never creates a state entry, so these states prove the firewall passed the traffic and the fault lay beyond it, in a service that was not listening. RED-02 at 0/0 B is the contrast. The greyed row is the legacy any-to-any rule, since deleted.

4. Review and recertification

- **Cadence:** the full rulebase is reviewed every **6 months** (PCI-DSS 1.1.7 discipline), or after any significant architecture change (new VLAN, new server role).
- **What a review checks:** each rule still has a valid justification; no shadowed/redundant/stale rules; least-privilege still holds; logging still appropriate; the register matches the live config.
- **Evidence:** review dates and outcomes are appended to the change control log.
- **Next review due:** 2027-02-15.

5. How to read a rule in this register (worked example: MGMT-01)

MGMT-01, Source MANAGEMENT net, Destination This Firewall, Service PFSENSE_MGMT (443/80/22), Action Pass, Log Yes.

- It only matches traffic **from** the management VLAN **to** pfSense itself, and only on the admin service ports. A host on any other VLAN, or aimed at any other destination, does not match it.
- It is logged because administrative access to the firewall is privileged activity that must be attributable.
- Its justification (anti-lockout) explains *why* it must be the first rule: if a later change removes the broad "allow" rule, this one guarantees the administrator can still reach pfSense to fix any mistake.
- Its controls (AC-17 remote access, CM-3 change control, AU-2 auditable events) are the standards language an auditor maps it to.

That is the level of "why" every rule in this register carries.

6. Known hardening opportunities (consciously accepted for now)

Recording a known weakness, the reason it is accepted, and the planned fix is itself a control (NIST CA-5, Plan of Action and Milestones). It shows a risk was identified and decided on deliberately, not missed.

ID	Observation	Risk	Current decision	Planned hardening	Trigger
H-01	MGMT-08 (RDP/WinRM to the DC) uses source = MANAGEMENT subnet , not specific hosts. Any device on VLAN 10 can <i>reach</i> the DC admin ports (network-location trust).	A rogue/compromised device placed on VLAN 10 could reach admin services. Mitigated by: (a) the DC still requires valid credentials + Remote Desktop Users membership to log in; (b) VLAN 10 is physically/administratively controlled.	Accepted while the lab has a single controlled admin laptop.	Create an ADMIN_HOSTS alias (specific admin/PAW IPs) and change MGMT-08 source to it; add MFA on the DC; move admin to a PAW.	When a second machine joins VLAN 10, or the PAW is built (see docs/12).
H-02	The TAILSCALE interface has broad rules: any tailnet device → pfSense admin; SSH to any destination; and <code>192.168.0.0/16 → any</code> (reach every VLAN). Remote access therefore bypasses the internal VLAN segmentation.	A compromised or shared tailnet device would have full lab reach, including the RedTeam VLAN.	Accepted , Tailscale is WireGuard-authenticated and the tailnet holds only 4 owner-controlled devices (pfSense subnet router <code>noble-home</code> ; admin laptop <code>noble-host 100.118.195.0</code> ; <code>lab-devops-svc01</code> ; <code>iphone182</code>). It is the sole remote-admin path, so tightening it hastily risks lockout.	Scope the pfSense-admin rule to the admin device (<code>noble-host</code>); optionally exclude RedTeam (VLAN 30) from Tailscale reach; prefer Tailscale ACLs (admin console) for device-level control. Physical VLAN-10 access (MGMT-01) is the fallback.	When convenient, and before adding any non-owner device to the tailnet.

| H-03 | **SIEM01** carries a local account `ansible` with full `sudo`, created when the machine was the monitoring host managed by the Ansible controller. **ANS01** no longer exists, so nothing uses it. | A dormant administrative account on the host that stores all security evidence. Unowned credentials are not attributable, and an account nobody uses is an account nobody watches. | **Locked, not deleted.** `passwd -l ansible` disables password authentication. The account is retained because the controller is being rebuilt and will need one. | Review when **ANS01** returns: either re-purpose it with a documented owner and scoped `sudo` rules, or delete it. Note that `passwd -l` does not block key-based SSH, so confirm `~ansible/.ssh/authorized_keys` is empty or remove it. | When **ANS01** is rebuilt, or at the next access review. |

H-02 was demonstrated, not merely theorised, on 2026-09-04. While the containment tests were proving that VLAN 30 cannot reach VLAN 50 or the management plane, an SSH session from the administrator's laptop to KALI01 on `192.168.30.2` was open and working. Both facts are true at once, and they are not contradictory: the SSH traffic arrived through the Tailscale subnet router rather than across a VLAN boundary, so pfSense never evaluated it against the REDTEAM interface rules. The documented rulebase governs the segment; it does not govern the overlay that terminates inside it.

This is the single most useful finding of the exercise. A rulebase can be individually correct on every interface and still not describe the reachability of the environment, because an authenticated overlay reaches past it. The honest statement of the control is therefore narrower than it first appears: *VLAN 30 is contained from the lab's routed paths, and remote-administration reachability is governed separately by Tailscale device membership.* Closing the gap means

either scoping the Tailscale rules per the planned hardening above, or moving that control into Tailscale ACLs where device identity, rather than network location, is the deciding factor. The second is the Zero Trust answer (NIST SP 800-207) and is the intended direction.

This register grows as other rules or segments are found to have similar location-based trust that could later be tightened to host- or identity-based trust (the Zero Trust direction, NIST SP 800-207).